

# HAT-P-32b

R-0050

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-32\_171109 · created 2026-07-09T08:02:47+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458066.8625 +/- 0.0024 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.001 +/- 0.054                 |
| Transit depth (Rp/Rs)^2                           | 0.00014 +/- 0.013 [%]           |
| Orbital Inclination (inc)                         | 84.82 +/- 1.72                  |
| Airmass coefficient 1 (a1)                        | 0.83 +/- 0.095                  |
| Airmass coefficient 2 (a2)                        | 0.17 +/- 0.14                   |
| Scatter in the residuals of the lightcurve fit is | 11.95 %                         |
| Best Comparison Star                              | #2 - [609, 432]                 |
| Optimal Aperture                                  | 3.54                            |
| Optimal Annulus                                   | 7.05                            |
| Transit Duration (day)                            | 0.1076 +/- 0.0084               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.001 ± 0.054 vs 0.1489 (deviation 2.74σ) |
| Duration fit vs published | 0.1076 d vs 0.1304 d (deviation 2.72σ)    |
| Mid-time O–C              | 2.3 min                                   |
| Depth SNR                 | 0.0                                       |
| Residual scatter          | 11.95 %                                   |

### Light curve

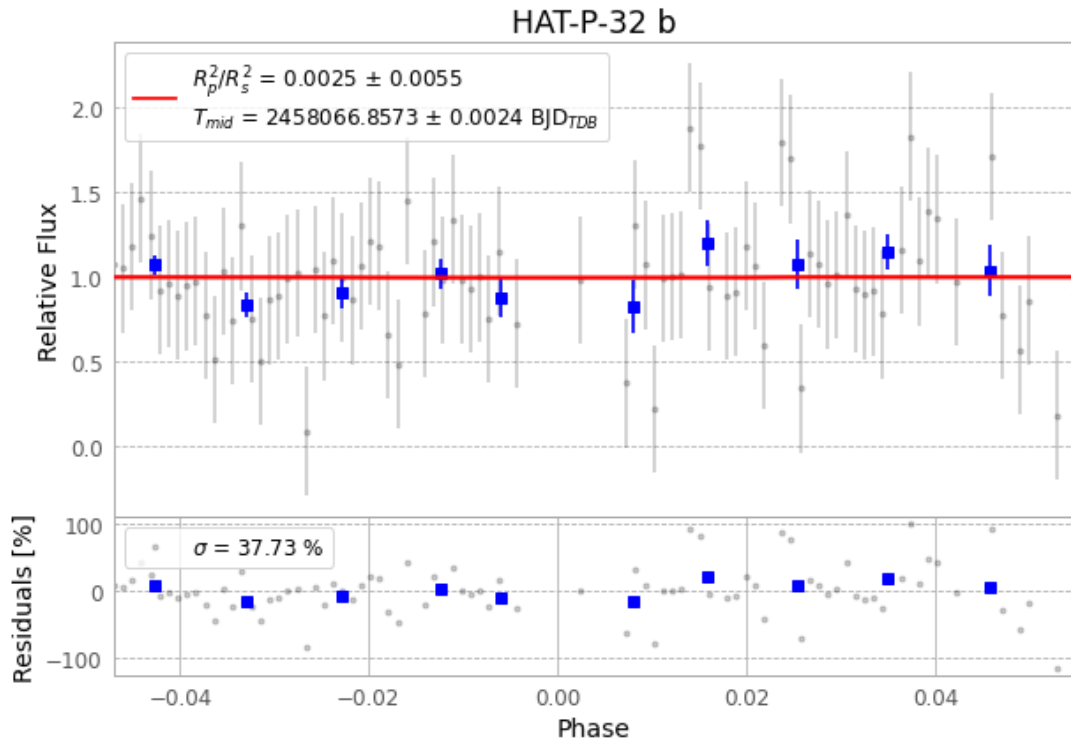


Figure 1 — FinalLightCurve\_HAT-P-32 b\_2017-11-09.png (SHA-256 1c9365575d9830cc...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 309], 2 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 80193e0a5e0cb1e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

## Data provenance

108 FITS frames (71 MB), HATP-32171109060012.FITS → HATP-32171109112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171109.log (8c046de1c360...), FinalLightCurve\_HAT-P-32 b\_2017-11-09.png (1c9365575d98...), FinalLightCurve\_HAT-P-32 b\_2017-11-08.png (7c59291b12db...), FinalLightCurve\_HAT-P-32 b\_2017-11-08.csv (3dd4845b9f14...), FinalLightCurve\_HAT-P-32 b\_2017-11-09.csv (96f618517904...)

## Compute resources

Wall time 163.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)